Chandrasekhar M V

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Accomplished Scientist/Engineer in Space Navigation with 15+ years of specialized experience in Positioning, Navigation, and Timing (PNT) technology and Onboard Satellite Positioning systems, built at ISRO across the full mission lifecycle — from concept design through development, testing, and operational deployment.

Proven leadership in directing multidisciplinary teams responsible for both ground-segment and onboard systems supporting GNSS broadcast services and spacecraft navigation. Combines deep technical expertise with strong techno-managerial acumen, having held project leadership roles in the design, development, and realization of navigation systems for satellite missions.

Held a rare, end-to-end skill set spanning from PNT to GNSS applications (Onboard Navigation). This dual strength across service-side infrastructure and application-layer development positions me uniquely to lead both foundational GNSS system builds and the downstream applications that depend on them.

Designation

Current Designation: Scientist / Engineer SF,
U. R. Rao Satellite Centre -ISRO, Bengaluru.
Work experience: 15 years, 08/2011 to till date

Project Roles

Deputy Project Director: Navigation systems (Onboard & Ground),

- Three High Resolution Optical Imaging Satellites
- One Stereographic mapping satellite

Project Manager: Satellite Positioning System

- Human space Program
- High resolution satellites constellation
- Micro satellites

Academics

Bachelor of Technology (B.Tech): Aeronautical Engineer – 2009, IARE, Hyderabad

Techinical Papers: 8

Highlights / Skill

1. Mission realization responsibilities and project management roles for navigation systems
2. PNT Techno – Managerial Lead
3. PNT- Ground Navigation Software, Systems Architecture, Design & Development Lead
4. Expertise in PNT Simulations & ground segment systems
5. Expertise in space navigation systems for Human Space Missions
6. Spaceborne GNSS systems expertise
7. Space situational awareness ground segment expertise
8. SSA data processing, objects identification and cataloguing algorithms expertise
9. Systems Testing, Validation & Qualification
10. Ground Orbit Determination knowledge
11. STK Tool expertise
12. Redhat Linux Operating systems

PNT

1. Pseudo range measurements with propagation errors simulation activities to assess the system design for preliminary and critical design reviews.

2. Analysis and development of navigation algorithms for the generation of broadcast navigation parameters which makes the core of the PNT services

3. Ground stations measurmnets processing and estimation algorithms for real time orbit and clock assessment.

4. PNT Ground navigation software and ground stations interfaces and data flow realization, deployment and operationalisation for everyday broadcast services

5. End to end payload testing & qualification activities

6. Systems lifecycle documentation, qualification through review committees and Test & Evaluations.

7. Short broadcast messaging servies for early warning and information dissimination systems

SSA

8. Space Resident object monitoring and caltaloguing using Ground sensors and OT data

9. Ground algorithms and software to process, Initial -OD, Object Identification, Catalogging & scheduling of space objects tracking.

Onboard Positioning & Timing Systems

10. Project navigatio & timing requirements assessment, specification derivation, Interfaces identification and Spacecraft mission realization activities

11. Navigation algorithms and software's development for the project deliverables

12. Anti-spoofing technology development activities.

13. Onboard software life cycle documentation

14. Unit level testing, validation and performance qualification of onboard GNSS systems

15. Systems qualification at environment test facilities and at dynamic test conditions

16. Performance qualification of onboard systems in closed Loop OILS and HILS testing

17. Mission phase readiness, Sequence of operations and POI activities

18. Ground based GNSS Orbit Determination activities

19. Qualifying the system through various requirements, design and test review committees.

Others

20. Futuristic X-ray Pulsar based navigation studies